

Question Number	Answer	Additional Guidance	Mark
4(a)	A description that includes two of the following points: • find a suitable {mass / age/species} of larvae (1) • find a suitable temperature for respiration (1) • find suitable method to measure the oxygen consumption (1) • suitable timescale to collect data (1)	A method to provide quantitative results Ignore amount/number of larvae Ignore comments about meat or liver Ignore respiration unqualified	2

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4(b)	An answer that includes nine of the following points: • clear statement of the dependent variable e.g. oxygen consumption per unit time (1) • use of a respirometer (1) • use of {same / stated} mass of larvae (1) • use of soda lime to absorb carbon dioxide (1) • detail of measuring dependent variable (1) • description of calculation of rate of gas consumption or production per unit mass (1) • Repeat with the other tissue (1) • two variables that need to be controlled (1) • method of control of one named variable (1) • repeat the whole experiment to calculate mean and {SD / error bars} (1)	Ignore amount/RQ Accept measure the distance (travelled) per unit time Accept respirometer described accept labelled diagram Ignore number of larvae/weight Accept sodium or potassium hydroxide but reject sodium hydrogen carbonate eg use of capillary tube or gas syringe Accept temperature / age / species of larvae / mass of meat /size of cube / species of meat Ignore light intensity/pH / volume of soda lime/weight Accept range bars	9

Question Number	Answer	Additional Guidance	Mark
4(c)	An answer that includes the following points: - table for collecting raw data (about liver and muscle) with headings and units with means calculated from repeats (1) - bar graph format with labelled axes (1) - use of a named statistical test for difference (1)	Not rate of... accept description of mean calculated in text or mean on one graph label if not in a table column eg t test U test Z test	3

Question Number	Answer	Additional Guidance	Mark
4(d)	An answer that includes two of the following points: - difficult to measure (small) distances along capillary tube with precision (1) {difficult to control age of larvae / respiration rate may change with changes in development of larvae} (1) - a stated difficulty with the apparatus (1)	Ignore refs to anaerobic/aerobic respiration MP1 accept (small) volumes of {gas/oxygen/carbon dioxide} MP1 ignore gas consumed or released Accept gas leakage/spent soda lime/death of some larvae/ temperature change due to respiration generating heat in the apparatus	2

Total for question 4 = 16

